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EXAMINING CPU INTERCONNECT BUSES IN A MULTIPROCESSOR SYSTEM

Abstract

A DMA controller of a peripheral switch is programmed to transfer memory contents to a switch memory including an instruction for a second processor. The instruction is sent from a second port of the switch coupled to a second peripheral lane of the second processor. Then, the contents are transferred to the switch memory using a first port of the switch coupled to a first peripheral lane of the first processor. The memory contents include first data transferred by the second processor to the first processor using a central processing unit (CPU) bus connection in response to the instruction.

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Background/Summary

BACKGROUND

[0001] Multiprocessor computer systems with multiple central processing units (CPUs) promise to increase computing power while remaining compatible with existing networks and operating systems. The server or desktop market segments often employ multiprocessor architectures, with each processor supporting multiple internal cores and various external communication buses. Interprocessor communication is performed using CPU interconnect buses that may be used in shared-memory multiprocessor systems to communicate between individual processors that can share a common memory space with each other, though each individual processor may have a non-uniform access to the shared memory with respect to the other individual processors. Furthermore, symmetric multiprocessor systems that utilize shared memory and a tightly-coupled symmetric multi-processor system that handles each processor in an identical manner have become a de-facto standard among processor manufacturers.

[0002] Thus, in a multiprocessor context, the distribution of computing tasks and access to the shared memory is coordinated and facilitated by the CPU interconnect buses that typically represent the highest throughput bandwidth and lowest latency inter-processor communication in such multiprocessor systems. However, an actual workload scalability, and hence, aggregate performance of any multiprocessor system, may increase slower than an actual number of individual processors used.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures.

[0004] FIG. 1 is a depiction of a test board for examining CPU interconnect buses in a multiprocessor system, in an example.

[0005] FIG. 2 is a depiction of a port in a test board, in an example.

[0006] FIG. 3A is a depiction of a quad processor test system configuration, in an example.

[0007] FIG. 3B is a depiction of a quad processor test system configuration, in an example.

[0008] FIG. 4 is a flowchart of a method of examining a multiprocessor system, in an example.

[0009] FIGS. 5A and 5B are a flowchart of a method of examining a multiprocessor system, in an example.

[0010] FIGS. 6A and 6B are a flowchart of a method of examining a multiprocessor system, in an example.

[0011] Corresponding numerals and symbols in the different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the disclosure and are not necessarily drawn to scale or perspective.

DETAILED DESCRIPTION

[0012] The following disclosure provides different examples for implementing different features. Specific examples of components and arrangements are depicted for descriptive clarity in the present disclosure. The present disclosure provides examples that are not intended to be limiting implementations of the subject matter disclosed herein.

[0013] Throughout this disclosure, a hyphenated form of a reference numeral refers to a specific instance of an element and the un-hyphenated form of the reference numeral refers to the element generically or collectively. Thus, as an example (not shown in the drawings), device “12-1” refers to an instance of a device class, which may be referred to collectively as devices “12” and any one of which may be referred to generically as a device “12”. In the figures and the description, like numerals are intended to represent like elements.

[0014] CPU performance has increased rapidly alongside the improvements in semiconductor

technology. The advancement has occurred with mobile processors but also with server systems, gaming systems, and other computer systems, that often employ multiprocessor architectures, with each processor supporting multiple internal cores and various external communication buses. CPU interconnect buses may be used in shared-memory multiprocessor systems to communicate between individual processors that can share a common memory space with each other, though each individual processor may have a non-uniform access to the shared memory with respect to the other individual processors. Furthermore, in some systems, symmetric multiprocessor systems use shared memory and a tightly-coupled symmetric multi-processor operating system may handle each processor in a similar manner.

[0015] In a multiprocessor context, the distribution of computing tasks and access to the shared memory may be coordinated and facilitated by the CPU interconnect buses that may provide high throughput bandwidth and low latency interprocessor communication in such multiprocessor systems. However, an actual workload scalability, and hence, aggregate performance of any multiprocessor system may increase slower than an actual number of individual processors used. The reasons for such reduced workload scalability can be complex and can depend on various internal and external factors, such as architectural capabilities of the hardware, as well as design of the software being executed on the hardware, such as the multiprocessor operating system used.

[0016] In operation of such multiprocessor architectures and components, the performance issues may arise or may be indicated within interprocessor communications that occur using the CPU interconnect buses, which link the individual processors together. Examining interprocessor communications can represent a fundamental technique for debugging and improving the overall workload scalability of multiprocessor systems. However, such examining may be highly complex and difficult due to the large volume of data and the high speed of the data transfers, along with the low latency of the CPU interconnect buses that are employed. This complexity and difficulty may increase as processor performance and capability increase, potentially resulting in greater losses of potential performance when coordination between individual processors is not sufficiently optimized to the workload.

[0017] The ability to debug and test CPU interconnect buses may be an important capability in the design, development, and evaluation of multiprocessor computer systems. For example, the ability to debug and test the CPU interconnect buses, as a part of examination, potentially while operating at or near full speed and at or near full capacity, which may be referred to as a 'saturation' condition of the interprocessor communications, may be important. As a particular example, when some or all CPU interconnect buses are operating in a saturation condition, certain errors may occur that adversely affect workload processing, but that are particularly difficult to detect and track. Therefore, the ability to saturate multiple CPU interconnect buses simultaneously and then examine the resulting behavior of each individual processor or other components in a multiprocessor system is an important technological issue.

[0018] As noted above, technological constraints have limited the ability of engineers to examine the operation of CPU interconnect buses due, for example, to throughput bandwidth and latency issues. However, as the performance of overall computer technology has advanced, the throughput bandwidth and latency of other bus systems used in computer systems has also improved. For example, the peripheral bus system used to connect external circuit boards and devices for direct communication with computer processors (e.g., Peripheral Component Interconnect Express (PCIe)) has also increased throughput bandwidth and reduced latency. The latest generations of PCIe, generations 3.0, 4.0, and 5.0, approach or exceed the throughput bandwidth of commonly used CPU interconnect buses.

[0019] Certain embodiments of this disclosure may use an external peripheral interconnect switch having multiple ports, including individual ports directly coupled to respective peripheral interconnect slots associated with respective individual processors in the multiprocessor system. Certain embodiments may allow examining of interprocessor communication traces at a switch

memory associated with the peripheral interconnect switch, while individual processors are not loaded with memory transfer operations to the switch memory. Certain embodiments may thus generate an instruction related to a transaction for a multiprocessor operating system executing on the multiprocessor system, and then examine interprocessor communications in the switch memory that result from the instruction. Certain embodiments may examine the interprocessor communication traces while the multiprocessor system is responding to the instruction.

[0020] Turning now to the drawings, FIG. 1 is a depiction of a test board **120** for examining CPU interconnect buses in a multiprocessor system. As used herein, “examining” can include or be associated with monitoring, testing, activating, tracing, tracking, recording, analyzing, debugging, capturing, triggering, calculating, outputting, among other related operations. In FIG. 1, test board **120** may represent an electronic device such as a printed circuit board (PCB) or similar type of circuit. As shown, test board **120** is populated with a switch **122** that is configured to switch among peripheral interconnect bus ports, referred to herein as switch ports or simply ports **124**, while switch **122** may be referred to as a bus switch, a peripheral interconnect switch, a peripheral switch, among other similar terms. In particular implementations, switch **122** represents a PCI Express (PCIe) switch. In various implementations, switch **122** supports PCIe devices and interfaces. As shown, switch **122** can include N number of ports, labeled as ports **124-1**, **124-2**, **124-3**, **124-4**, **124-5**, **124-6**, **124-7**, **128-8**, up to **124-N**.

[0021] Also shown included with switch **122** in FIG. 1 is a switch buffer **130** that may represent a common memory accessible to ports **124** included in switch **122**. Switch **122** is also shown including one or more direct memory access (DMA) controllers **140** that can be configured to transfer data from different memories without involvement of another processor, such as test data **306** from a memory **304** to switch buffer **130** (see FIGS. 3A and 3B). It is noted that switch **122** may further include components for performing data processing, such as for controlling ports **124**, accessing switch buffer **130**, among other tasks. Accordingly, switch **122** includes a controller **150** and a firmware **152** that is accessible to and can store instructions executable by controller **150**. Controller **150** can include functionality for performing data processing, such as by executing the instructions stored in firmware **152**, including instructions to control DMA controller(s) **140**. For example, controller **150** can send a first memory transfer instruction to a DMA controller **140** to configure DMA controller **140** to transfer test data **306** from a memory **304** to switch buffer **130**, and can send a second memory transfer instruction to DMA controller **140** to initiate the transfer of test data **306**, after which DMA controller **140** can independently perform the transfer.

[0022] FIG. 2 is a depiction of port **124** in test board **120**. In FIG. 2, further details of port **124** are shown. Port **124** may include a packet generator **210** that may format data for outputting and receiving data in a desired packet format. Port **124** may further include executable instructions **214** that can be executed at switch **122**, including a data pattern **214-1**, a packet payload **214-2**, read/write request **214-3**, and memory transfer **214-4**. Thus instructions **214** may configure port **124** and switch **122** to perform operations on a peripheral interconnect bus coupled to switch **122**, such as sending instructions, receiving instructions, accessing memory (read and write) locations, as well as programming and initiating DMA transfers from a memory endpoint, such a memory device accessible via and coupled to port **124**. Also as shown, port **124** includes a write queue **216** and a read queue **218** that may respectively send and receive data, such as packets handled by packet generator **210**, to and from the peripheral interconnect bus. Write queue **216** and read queue **218** may be configured to access switch buffer **130** that can support multiple ports **124**.

[0023] Referring now to FIG. 3A, a quad processor test system configuration **300**, or simply configuration **300**, is depicted schematically. Configuration **300** in FIG. 3A includes a CPU interconnect bus **310** that provides high speed bidirectional communication among processors **302**, specifically first processor **302-1**, second processor **302-2**, third processor **302-3**, and fourth processor **302-4**. Although 4 processors are shown in configuration **300** for descriptive clarity, fewer or more processors **302** can be used in different implementations.

[0024] In FIG. 3A, CPU interconnect bus **310** may represent physical interconnections between processors **302** and is shown having logical connections **312** established thereon that can be configured for unidirectional or bidirectional data transfer, also referred to as data transmission, between individual processors **302**. Specifically, logical connection **312-1** establishes data transfer between first processor **302-1** and second processor **302-2**, logical connection **312-2** establishes data transfer between second processor **302-2** and third processor **302-3**, logical connection **312-3** establishes data transfer between third processor **302-3** and fourth processor **302-4**, and logical connection **312-4** establishes data transfer between fourth processor **302-4** and first processor **302-1**. It is noted that logical connections **312** may be used simultaneously and bidirectionally in various implementations.

[0025] In FIG. 3A, processors **302** are shown including multiple peripheral bus lanes **308** that support the peripheral interconnect bus that switch **122** and ports **124** support. Accordingly, each processor **302** in configuration **300** is shown connected to two individual ports **124** of switch **122** at peripheral bus lanes **308**. Specifically, first processor **302-1** is coupled to ports **124-1** and **124-2**, second processor **302-2** is coupled to ports **124-3** and **124-4**, third processor **302-3** is coupled to ports **124-5** and **124-6**, and fourth processor **302-4** is coupled to ports **124-7** and **124-8**.

[0026] In configuration **300** of FIG. 3A, processors **302** are further connected to a respective memory **304** that is shown storing test data **306**. Specifically, first processor **302-1** is connected to memory **304-1** storing test data **306-1**, second processor **302-2** is connected to memory **304-2** storing test data **306-2**, third processor **302-3** is connected to memory **304-3** storing test data **306-3**, and fourth processor **302-4** is connected to memory **304-4** storing test data **306-4**.

[0027] In operation of configuration **300**, port **124** can be used to send and receive data and instructions to processors **302** to perform various examination procedures, such as by using logical connections **312** to transmit data and examine the CPU interconnect buses among processors **302**, including recording the results to analyze interprocessor communication traces, for example. Accordingly, test data **306** can be written via port **124** or can be written by a respective processor **302**. Test data **306** can also include results or processor output to an instruction received by processor **302**. In the multiprocessor system shown, an instruction executed by one processor **302** can result in test data **306** being generated and locally stored by another processor **302** in its respective memory **304**. Various other types of data transfer and instructions can be performed for examining interprocessor communication using configuration **300**, as will be described in further detail.

[0028] Because two ports **124** of switch **122** are coupled to each processor **302** in configuration **300**, two logical connections **312** can be examined per processor **302**, since each port **124** can provide sufficient or maximum data throughput commensurate with one logical connection **312**.

[0029] Further, using configuration **300**, each logical connection **312** can be examined in full duplex or full bidirectional operation, since one port **124** is available for each endpoint of logical connection **312**. In this manner, data can be streamed to processor **302** or memory **304** from one port **124**, transmitted to another processor **302** or memory **304**, and received by another port **124**. Since ports **124** are included with switch **122**, data or results involved with such an examination can be recorded in switch buffer **130** for post-examination analysis and evaluation, such as to determine errors or latency from interprocessor communication traces. In some cases, a result of the examination can be trace data that confirms the stability and proper operation of a multiprocessor system being examined while operating at maximum data throughput rates on the connected CPU interconnect buses, which can include 2, 4, 6, or 8 ports **124** operating for bidirectional data transfer with simultaneous operation.

[0030] In various implementations, because ports **124** can receive and transmit data at a maximum data rate supported by CPU interconnect bus **310**, one or more or all logical connections **312** can be saturated or flooded with data at the maximum data rate. In particular test cases, four logical connections **312** can be simultaneously saturated with data in a bidirectional manner using

configuration **300**. In this manner, the operation and behavior of processors **302** under saturation conditions of CPU interconnect bus **310** can be reliably examined, which is desirable. In particular performance issues at performance limits of operation can be identified and resolved, so that more scalable overall performance of the multiprocessor computer system can be realized.

[0031] Referring now to FIG. **3B**, a quad processor test system configuration **301**, or simply configuration **301**, is depicted schematically. Configuration **301** in FIG. **3B** includes a CPU interconnect bus **310** that provides high speed bidirectional communication among processors **302**, specifically first processor **302-1**, second processor **302-2**, third processor **302-3**, and fourth processor **302-4**. As shown configuration **301** is substantially similar to configuration **300** described above with respect to FIG. **3A**.

[0032] In configuration **301** of FIG. **3B**, instead of 2 ports **124** per processor **302** as in configuration **300** of FIG. **3A**, 3 ports **124** per processor **302** are coupled via respective peripheral bus lanes **308** provided by processor **302**. Accordingly, each processor **302** in configuration **301** is shown connected to three individual ports **124** of switch **122** at peripheral bus lanes **308**.

Specifically, first processor **302-1** is coupled to ports **124-1**, **124-2**, and **124-3**, second processor **302-2** is coupled to ports **124-4**, **124-5**, and **124-6**, third processor **302-3** is coupled to ports **124-7**, **124-8**, and **124-9**, and fourth processor **302-4** is coupled to ports **124-10**, **124-11**, and **124-12**.

[0033] In FIG. **3B**, CPU interconnect bus **310** may represent physical interconnections between processors **302** and is shown having logical connections **312** established thereon that can be configured for unidirectional or bidirectional data transfer, also referred to as data transmission, between individual processors **302**. Specifically, logical connection **312-1** establishes data transfer between first processor **302-1** and second processor **302-2**, logical connection **312-2** establishes data transfer between second processor **302-2** and third processor **302-3**, logical connection **312-3** establishes data transfer between third processor **302-3** and fourth processor **302-4**, logical connection **312-4** establishes data transfer between fourth processor **302-4** and first processor **302-1**, logical connection **312-5** establishes data transfer between first processor **302-1** and third processor **302-3**, and logical connection **312-6** establishes data transfer between second processor **302-2** and fourth processor **302-4**. It is noted that logical connections **312** may be used simultaneously and bidirectionally in various implementations.

[0034] In operation of configuration **301** for testing processors **302**, each processor **302** can establish 3 concurrent, bidirectional logical connections **312** to each other processor in the quad processor system shown in FIG. **3B**. Using three ports **124** per processor **302**, switch **122** can flood three logical connections **312** to each of the other three processors **302**. Further, the same type of configuration of logical connections **312** can be concurrently established for each of the four processors **302** in configuration **301**. In this manner, six logical connections **312** can be simultaneously saturated with data being transferred bidirectionally at full data throughput rates between six pairs of processors **302** in configuration **301**, which is desirable. In addition to achieving the saturation condition for CPU interconnect bus **310**, interprocessor communication traces and all data transferred by CPU interconnect bus **310** can be collected by switch **122** in switch buffer **130**.

[0035] Depending on an N number of ports **124** available at switch **122**, one or more switches **122** may be used in larger or other configurations where greater values of N number of ports **124** is desired or higher data rates are desired. For example, a certain ratio of ports **124** to logical connections **312** may be determined by maximum data transfer rates of peripheral bus lanes **308**, CPU interconnect bus **310**, or both, in different implementations.

[0036] FIG. **4** is a flowchart of a method **400** of examining a multiprocessor system. Method **400** may be performed using configuration **300** depicted in and as described above with respect to FIG. **3A**. Various operations in method **400** may be rearranged or omitted in various cases.

[0037] Method **400** begins at step **402** by sending an instruction to a second processor from a second port of a peripheral interconnect switch coupled to a second peripheral lane of the second

processor, the instruction being associated with contents of a first memory associated with a first processor included in a multiprocessor system with the second processor. In response to sending the instruction and using a direct memory access (DMA) controller of the peripheral interconnect switch, at step **404**, at least some of the contents of the first memory are transferred to a switch memory associated with the peripheral interconnect switch using a first port of the peripheral interconnect switch coupled to a first peripheral lane of the first processor, where the contents of the first memory include first data transferred by the second processor to the first processor using a first CPU bus connection in response to the instruction. The first data can be transferred using the first CPU bus connection at a maximum data transfer rate supported by the multiprocessor system. The contents of the first memory that are transferred to the switch memory can be less than all contents of the first memory. At step **406**, during transfer of the contents by the DMA controller, at least some of the first data stored in the switch memory are analyzed. Analyzing at least some of the first data stored in the switch memory can include analyzing latency of the first data, or analyzing errors in the first data, or both.

[0038] FIGS. 5A and 5B are flowcharts of a method **500** of examining a multiprocessor system. Method **500** is split into two portions **500-1** and **500-2** in FIGS. 5A and 5B respectively, and may be performed using configuration **301** depicted in and as described above with respect to FIG. 3B. Various operations in method **500** may be rearranged or omitted in various cases. Method **500** may be performed by a test board including a bus switch having multiple ports and a common buffer, where the bus switch supports a peripheral interconnect bus.

[0039] Method **500-1** in FIG. 5A begins at step **502** by instructing a first direct memory access (DMA) controller of the bus switch to transmit, from a first port of the bus switch, first data to a second memory associated with a second processor included with a computer system, where a first peripheral lane, a second peripheral lane, and a third peripheral lane associated with a first processor included in the computer system are respectively connected to the first port, a second port, and a third port of the bus switch, and where the first processor is associated with a first memory and a fourth peripheral lane associated with the second processor is connected to a fourth port of the bus switch.

[0040] At step **504**, a second DMA controller of the bus switch is instructed to transmit, from the second port, second data to a third memory associated with a third processor included in the computer system, wherein a fifth peripheral lane associated with the third processor is connected to a fifth port of the bus switch. At step **506**, a third DMA controller of the bus switch is instructed to transmit, from the third port, third data to a fourth memory associated with a fourth processor included in the computer system, wherein a sixth peripheral lane associated with the fourth processor included in the computer system is connected to a sixth port of the bus switch. At step **508**, a transfer of the first data, the second data, and the third data respectively and concurrently is initiated, where the first data is transmitted using a first CPU bus connection to the second memory, the second data is transmitted using a second CPU bus connection to the third memory, and the third data is transmitted using a third CPU bus connection to the fourth memory. At step **510**, the first data are received at the fourth port using the fourth peripheral lane, the second data at the fifth port are received using the fifth peripheral lane, and the third data at the sixth port are received using the sixth peripheral lane. After step **510** method **500** proceeds to method **500-2** in FIG. 5B.

[0041] Method **500-2** in FIG. 5B begins at step **512** by writing the first data, the second data, and the third data to the common buffer from the fourth port, the fifth port, and the sixth port respectively. At step **514**, the first data, the second data, and the third data stored in the common buffer are analyzed, including respectively analyzing the first data, the second data, and the third data for errors. At step **516**, a fourth DMA controller of the bus switch is programmed to transmit, from the fourth port, fourth data to the first memory, a fifth DMA controller of the bus switch is programmed to transmit, from the fifth port, fifth data to the first memory, and a sixth DMA controller of the bus switch is programmed to transmit, from the sixth port, sixth data to the first

memory. At step **518**, the transfer of the fourth data, the fifth data, and the sixth data respectively and concurrently is initiated, where the fourth data is transmitted using the first CPU bus connection to the first memory, the fifth data is transmitted using the second CPU bus connection to the first memory, and the sixth data is transmitted using the third CPU bus connection to the first memory.

[0042] FIGS. **6A** and **6B** are flowcharts of a method **600** of examining a multiprocessor system. Method **600** is split into two portions **600-1** and **600-2** in FIGS. **6A** and **6B**, respectively, and may be performed using configuration **301** depicted in and as described above with respect to FIG. **3B**. Various operations in method **600** may be rearranged or omitted in various cases.

[0043] Method **600-1** in FIG. **6A** begins at step **602** by instruct a first direct memory access (DMA) controller of the bus switch to transmit, from a second memory associated with the second processor, first data to the common buffer, the first data generated by the second processor in response to a first instruction. At step **604**, a second DMA controller of the bus switch is instructed to transmit, from a third memory associated with the third processor, second data to the common buffer, the second data generated by the third processor in response to a second instruction.

[0044] At step **606**, a third DMA controller of the bus switch is instructed to transmit, from a fourth memory associated with the fourth processor, third data to the common buffer, the third data generated by the fourth processor in response to a third instruction. At step **608**, the first instruction is sent from the first port to the second processor, the second instruction from the second port to the third processor, and the third instruction from the third port to the fourth processor. At step **610**, the first data, the second data, and the third data are accessed in the common buffer, where the first data is transmitted using a first central processing unit (CPU) bus connection from the second memory, the second data is transmitted using a second CPU bus connection from the third memory, and the third data is transmitted using a third CPU bus connection from the fourth memory. After step **610** method **600** proceeds to method **600-2** in FIG. **6B**.

[0045] Method **600-2** in FIG. **6B** begins at step **612** by receiving the first data at the first port using the first peripheral lane, receiving the second data at the second port using the second peripheral lane and receiving the third data at the third port using the third peripheral lane. At step **614**, the first data, the second data, and the third data stored in the common buffer are analyzed, including respectively analyzing the first data, the second data, and the third data for errors or latency.

[0046] As disclosed herein, a DMA controller of a peripheral switch is programmed to transfer memory contents to a switch memory including an instruction for a second processor. The instruction is sent from a second port of the switch coupled to a second peripheral lane of the second processor. Then, the contents are transferred to the switch memory using a first port of the switch coupled to a first peripheral lane of the first processor. The memory contents include first data transferred by the second processor to the first processor using a central processing unit (CPU) bus connection in response to the instruction.

[0047] The foregoing outlines features of several examples so that those skilled in the art may better understand the aspects of the present disclosure. Various modifications and combinations of the illustrative examples, as well as other examples, are contemplated in the description.

Claims

1. A method, comprising: sending an instruction to a second processor from a second port of a peripheral interconnect switch coupled to a second peripheral lane of the second processor, the instruction being associated with contents of a first memory associated with a first processor included in a multiprocessor system with the second processor; and transferring, in response to sending the instruction and using a direct memory access (DMA) controller of the peripheral interconnect switch, at least some of the contents of the first memory to a switch memory associated with the peripheral interconnect switch using a first port of the peripheral interconnect

switch coupled to a first peripheral lane of the first processor, wherein the contents of the first memory include first data transferred by the second processor to the first processor using a first central processing unit (CPU) bus connection in response to the instruction.

2. The method of claim 1, further comprising: analyzing, during the transferring by the DMA controller, at least some of the first data stored in the switch memory.

3. The method of claim 2, wherein analyzing at least some of the first data stored in the switch memory includes at least one of: analyzing latency of the first data, or analyzing errors in the first data.

4. The method of claim 1, wherein the first data are transferred using the first CPU bus connection at a maximum data transfer rate supported by the multiprocessor system.

5. The method of claim 1, wherein the contents of the first memory that are transferred to the switch memory are less than all contents of the first memory.

6. A device, comprising: a test board including a bus switch having multiple ports and a common buffer, wherein the bus switch supports a peripheral interconnect bus, and wherein the test board is configured to: instruct a first direct memory access (DMA) controller of the bus switch to transmit, from a first port of the bus switch, first data to a second memory associated with a second processor included with a computer system, wherein a first peripheral lane, a second peripheral lane, and a third peripheral lane associated with a first processor included in the computer system are respectively connected to the first port, a second port, and a third port of the bus switch, and wherein the first processor is associated with a first memory and a fourth peripheral lane associated with the second processor is connected to a fourth port of the bus switch; instruct a second DMA controller of the bus switch to transmit, from the second port, second data to a third memory associated with a third processor included in the computer system, wherein a fifth peripheral lane associated with the third processor is connected to a fifth port of the bus switch; instruct a third DMA controller of the bus switch to transmit, from the third port, third data to a fourth memory associated with a fourth processor included in the computer system, wherein a sixth peripheral lane associated with the fourth processor included in the computer system is connected to a sixth port of the bus switch; and initiate a transfer of the first data, the second data, and the third data respectively and concurrently, wherein the first data is transmitted using a first central processing unit (CPU) bus connection to the second memory, the second data is transmitted using a second CPU bus connection to the third memory, and the third data is transmitted using a third CPU bus connection to the fourth memory.

7. The device of claim 6, wherein the test board is further configured to: receive the first data at the fourth port using the fourth peripheral lane; receive the second data at the fifth port using the fifth peripheral lane; and receive the third data at the sixth port using the sixth peripheral lane.

8. The device of claim 7, wherein the test board is further configured to: write the first data, the second data, and the third data to the common buffer from the fourth port, the fifth port, and the sixth port respectively.

9. The device of claim 8, wherein the test board is further configured to: analyze the first data, the second data, and the third data stored in the common buffer, including respectively analyzing the first data, the second data, and the third data for errors.

10. The device of claim 6, wherein: the first data are transmitted a maximum data transfer rate supported by the computer system; the second data are transmitted at the maximum data transfer rate; and the third data are transmitted at the maximum data transfer rate.

11. The device of claim 6, wherein: the first CPU bus connection communicates between the first processor and the second processor; the second CPU bus connection communicates between the first processor and the third processor; and the third CPU bus connection communicates between the first processor and the fourth processor.

12. The device of claim 6, wherein the test board is further configured to: instruct a fourth DMA controller of the bus switch to transmit, from the fourth port, fourth data to the first memory;

instruct a fifth DMA controller of the bus switch to transmit, from the fifth port, fifth data to the first memory; instruct a sixth DMA controller of the bus switch to transmit, from the sixth port, sixth data to the first memory; and initiate a transmission of the fourth data, the fifth data, and the sixth data respectively and concurrently, wherein the fourth data is transmitted using the first CPU bus connection to the first memory, the fifth data is transmitted using the second CPU bus connection to the first memory, and the sixth data is transmitted using the third CPU bus connection to the first memory.

13. The device of claim 12, wherein the first data, the second data, the third data, the fourth data, the fifth data, and the sixth data are transmitted concurrently.

14. A system, comprising: a computer system comprising a first processor, a second processor, a third processor, and a fourth processor; a test board including a bus switch having multiple ports and a common buffer, wherein the bus switch supports a peripheral interconnect bus; a first peripheral lane, a second peripheral lane, and a third peripheral lane associated with the first processor respectively connected to a first port, a second port, and a third port of the bus switch, wherein the first processor is associated with a first memory; a fourth peripheral lane associated with the second processor connected to a fourth port of the bus switch; a fifth peripheral lane associated with the third processor connected to a fifth port of the bus switch; and a sixth peripheral lane associated with the fourth processor connected to a sixth port of the bus switch.

15. The system of claim 14, wherein the test board is configured to: instruct a first direct memory access (DMA) controller of the bus switch to transmit, from a second memory associated with the second processor, first data to the common buffer, the first data generated by the second processor in response to a first instruction; instruct a second DMA controller of the bus switch to transmit, from a third memory associated with the third processor, second data to the common buffer, the second data generated by the third processor in response to a second instruction; instruct a third DMA controller of the bus switch to transmit, from a fourth memory associated with the fourth processor, third data to the common buffer, the third data generated by the fourth processor in response to a third instruction; send the first instruction from the first port to the second processor, the second instruction from the second port to the third processor, and the third instruction from the third port to the fourth processor; and access the first data, the second data, and the third data in the common buffer, wherein the first data is transmitted using a first central processing unit (CPU) bus connection from the second memory, the second data is transmitted using a second CPU bus connection from the third memory, and the third data is transmitted using a third CPU bus connection from the fourth memory.

16. The system of claim 15, wherein the test board is further configured to: receive the first data at the first port using the first peripheral lane; receive the second data at the second port using the second peripheral lane; and receive the third data at the third port using the third peripheral lane.

17. The system of claim 16, wherein the test board is further configured to: analyze the first data, the second data, and the third data stored in the common buffer for errors.

18. The system of claim 17, wherein the test board is further configured to: respectively analyze the first data, the second data, and the third data for latency.

19. The system of claim 15, wherein: the first data are transmitted at a maximum data transfer rate supported by the computer system; the second data are transmitted at the maximum data transfer rate; and the third data are transmitted at the maximum data transfer rate.

20. The system of claim 15, wherein: the first CPU bus connection communicates between the first processor and the second processor; the second CPU bus connection communicates between the first processor and the third processor; and the third CPU bus connection communicates between the first processor and the fourth processor.
